Solution

The rotational kinetic energy is, $KE = \frac{1}{2} I\omega^2$

The moment of inertia of the ring is, $I = MR^2$

$$I = 9 \times 3^2 = 9 \times 9 = 81\,\text{kg m}^2$$

The angular speed of the ring is,

$$\omega = 240\,\text{rpm} = \frac{240 \times 2\pi}{60}\,\text{rad s}^{-1}$$

$$KE = \frac{1}{2} \times 81 \times \left(\frac{240 \times 2\pi}{60}\right)^2 = \frac{1}{2} \times 81 \times (8\pi)^2$$

$$KE = \frac{1}{2} \times 81 \times 64 \times (\pi)^2 = 2592 \times (\pi)^2$$

$$KE \approx 25920\,\text{J} \quad \because (\pi)^2 \approx 10$$

$$KE = 25.920\,\text{kJ}$$

5.5.5 Power Delivered by Torque

Power delivered is the work done per unit time. If we differentiate the expression for work done with respect to time, we get the instantaneous power (P).

$$P = \frac{dw}{dt} = \tau \frac{d\theta}{dt} \quad \because (dw = \tau d\theta)$$

$$P = \tau\omega \qquad (5.54)$$

The analogous expression for instantaneous power delivered in translational motion is,

$$P = \vec{F} \cdot \vec{v}$$

5.5.6 Comparison of Translational and Rotational Quantities

Many quantities in rotational motion have expressions similar to that of translational motion. The rotational terms are compared with the translational equivalents in Table 5.4.

Table 5.4 Comparison of Translational and Rotational Quantities

S.No	Translational Motion	Rotational motion about a fixed axis
1	Displacement, x	Angular displacement, θ
2	Time, t	Time, t
3	Velocity, $v = \frac{dx}{dt}$	Angular velocity, $\omega = \frac{d\theta}{dt}$
4	Acceleration, $a = \frac{dv}{dt}$	Angular acceleration, $\alpha = \frac{d\omega}{dt}$
5	Mass, m	Moment of inertia, I
6	Force, F = ma	Torque, $\tau = I\,\alpha$
7	Linear momentum, p = mv	Angular momentum, L = Iω
8	Impulse, $F\,\Delta t = \Delta p$	Angular impulse, $\tau\Delta t = \Delta L$
9	Work done, w = F s	Work done, $w = \tau\,\theta$
10	Kinetic energy, $KE = \frac{1}{2} m v^2$	Kinetic energy, $KE = \frac{1}{2} I\omega^2$
11	Power, P = F v	Power, $P = \tau\omega$

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